SAMPLE — FICTIONAL DOCUMENT FOR PROTOTYPE DEMONSTRATION ONLY

SAMPLE REFINERY LTD. — INSPECTION DEPARTMENT

INSPECTION REPORT No. IR-2026-118

Date of inspection: 14-10-2026 Unit: Crude Distillation Unit

Equipment: Crude charge pump P-2105B (Criticality: C)

Inspected by: Condition Monitoring Engineer

1. Vibration readings (mm/s RMS)

Location	Horizontal	Vertical	Axial
DE bearing	6.4	4.1	3.2
NDE bearing	5.1	3.6	2.7

2. Bearing housing temperature

Location	Previous (14-09-2026)	Current
DE bearing	68 °C	86 °C
NDE bearing	65 °C	74 °C

3. Observations

a) Highest vibration 6.4 mm/s at DE horizontal. Spectrum shows peaks at bearing defect frequencies, indicating DE bearing wear.

b) DE bearing temperature rose by 18 °C since the last reading.

c) Oil level normal. Oil sample slightly darkened.

d) Coupling alignment not checked (pump in service).

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4. Recommendations

1) Change over to standby pump P-2105A after confirming its condition, and replace DE and NDE bearings of P-2105B.
2) Carry out laser alignment of pump and motor after bearing job.
3) Replace lubricating oil and send old oil for analysis.
4) Increase vibration monitoring of P-2105B to weekly until readings return to Zone B or better.

5. Estimated cost

Item	Amount (Rs)
Bearings (1 set, DE + NDE)	65,000
Laser alignment + labour	48,000
Lubricating oil + analysis	12,000
Total	1,25,000

Condition Monitoring Engineer (signed)

Area Maintenance Engineer (signed)